

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
df=pd.read_csv('insurance.csv')
```

```
df.head()
```

	age	sex	bmi	children	smoker	region	charges	
0	19.0	female	27.900	0.0	yes	southwest	16884.92400	
1	18.0	male	33.770	1.0	no	southeast	1725.55230	
2	28.0	male	33.000	3.0	no	southeast	4449.46200	
3	33.0	male	22.705	0.0	no	northwest	21984.47061	
4	32.0	male	28.880	0.0	no	northwest	3866.85520	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
df.tail()
```

	age	sex	bmi	children	smoker	region	charges	
1333	50.0	male	30.97	3.0	no	northwest	10600.5483	
1334	18.0	female	31.92	0.0	no	northeast	2205.9808	
1335	18.0	female	36.85	0.0	no	southeast	1629.8335	
1336	21.0	female	25.80	0.0	no	southwest	2007.9450	
1337	61.0	female	NaN	0.0	yes	northwest	29141.3603	

```
df.shape
```

```
(1338, 7)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 1338 entries, 0 to 1337  
Data columns (total 7 columns):  
 #   Column      Non-Null Count  Dtype    
---  -  
 0   age         1337 non-null   float64  
 1   sex         1338 non-null   object   
 2   bmi         1337 non-null   float64  
 3   children    1337 non-null   float64  
 4   smoker      1338 non-null   object   
 5   region      1338 non-null   object   
 6   charges     1338 non-null   float64  
dtypes: float64(4), object(3)  
memory usage: 73.3+ KB
```

```
df.isnull().sum()
```

```
      0  
age    1  
sex    0  
bmi    1  
children 1  
smoker  0  
region  0  
charges 0  
  
dtype: int64
```

```
df
```

	age	sex	bmi	children	smoker	region	charges	
0	19.0	female	27.900	0.0	yes	southwest	16884.92400	
1	18.0	male	33.770	1.0	no	southeast	1725.55230	
2	28.0	male	33.000	3.0	no	southeast	4449.46200	
3	33.0	male	22.705	0.0	no	northwest	21984.47061	
4	32.0	male	28.880	0.0	no	northwest	3866.85520	
...	...	...	...	...	...	...	...	
1333	50.0	male	30.970	3.0	no	northwest	10600.54830	
1334	18.0	female	31.920	0.0	no	northeast	2205.98080	
1335	18.0	female	36.850	0.0	no	southeast	1629.83350	
1336	21.0	female	25.800	0.0	no	southwest	2007.94500	
1337	61.0	female	NaN	0.0	yes	northwest	29141.36030	

1338 rows × 7 columns

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```

df=pd.read_csv('insurance.csv')

# Impute missing numerical values with the mean
df['age'] = df['age'].fillna(df['age'].mean())
df['bmi'] = df['bmi'].fillna(df['bmi'].mean())
df['children'] = df['children'].fillna(df['children'].mean())

# Define features
x=df[['age', 'sex', 'bmi', 'children', 'smoker', 'region']]
y=df['charges']

# Perform one-hot encoding on categorical columns in x
x = pd.get_dummies(x, columns=['sex', 'smoker', 'region'], drop_first=True)

```

```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
```

```
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
```

▼ LinearRegression ⓘ ?

LinearRegression()

```
y_pred=model.predict(x_test)
mse=mean_squared_error(y_test,y_pred)
r2=r2_score(y_test,y_pred)
print(mse)
print(r2)
```

```
33583187.323717855
0.7836814059540487
```

```
new_data = {
    'age': [30],
    'bmi': [25.0],
    'children': [1],
    'sex_male': [1], # 1 for male, 0 for female
    'smoker_yes': [0], # 1 for smoker, 0 for non-smoker
    'region_northwest': [0],
    'region_southeast': [1],
    'region_southwest': [0]
}
```

```
new = pd.DataFrame(new_data)
```

```
# Ensure column order matches the training data if needed
# For robustness, you could reindex columns based on x_train.columns
# For now, let's assume direct creation is sufficient if x_train is available
# Alternatively, if x is available:
# new = new[x.columns] # Uncomment if you want to enforce column order
```

```
predicted_count = model.predict(new)
print(predicted_count)
```

```
[ 3958.0433411]
```